

Normalized spectral support need not fill an interval

Disproof of Conjecture 00000003702

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Scope of the claim. Both language versions state that the limiting measure has support the interval $[0, 2]$. We disprove this literal equality. The usual statement that normalized Laplacian spectra are *contained* in $[0, 2]$ is valid and is not contradicted here. The source imposes no connectedness hypothesis. Our graphs have unbounded vertex count.

Actual graphs and normalized operator. For $m \geq 0$, let G_m be the disjoint union of $m + 1$ copies of K_2 . More explicitly its vertices are (c, i) with $c \in \{0, \dots, m\}$ and $i \in \{0, 1\}$; adjacency holds exactly when the copy labels agree and the endpoints differ. Every vertex has its opposite endpoint as its unique neighbor, so every degree is one and $|V(G_m)| = 2(m+1) \rightarrow \infty$. Consequently the genuine normalized Laplacian is

$$L_m = I - D^{-1/2} A_m D^{-1/2} = I - A_m, \quad (L_m f)(c, i) = f(c, i) - f(c, 1 - i).$$

The random-walk normalization $I - D^{-1} A_m$ gives the same operator because $D = I$.

Complete spectrum. On each edge the symmetric vector $(1, 1)$ has eigenvalue zero, and the antisymmetric vector $(1, -1)$ has eigenvalue two. They form a basis of that edge's two-dimensional function space. Putting these bases on the disjoint copies gives a complete $2(m + 1)$ -element eigenbasis of the actual graph operator. Its invertible coordinates on copy c are

$$a_c = \frac{f(c, 0) + f(c, 1)}{2}, \quad b_c = \frac{f(c, 0) - f(c, 1)}{2}, \quad f(c, i) = a_c + (-1)^i b_c.$$

Thus there are exactly $m + 1$ eigenvalues zero and $m + 1$ eigenvalues two, including multiplicities.

Empirical measures and weak limit. The ordinary normalized empirical spectral probability measure is

$$\mu_m = \frac{1}{2(m+1)} \sum_{c=0}^m (\delta_0 + \delta_2) = \frac{1}{2} \delta_0 + \frac{1}{2} \delta_2 =: \mu.$$

This is a constant sequence of actual empirical measures, so it converges weakly to μ ; equivalently every bounded continuous test function has identical integral against every μ_m and μ .

Failure of full support. The open neighborhood $U = (1/2, 3/2)$ of the point 1 has $\mu(U) = 0$, since it contains neither atom. By the standard neighborhood definition of topological support, 1 is absent from $\text{supp } \mu$, although $1 \in [0, 2]$. Indeed the support is exactly $\{0, 2\}$. Therefore the limiting support is not the full interval asserted in the conjecture. This false clause suffices for the disproof even though this particular sequence does converge.

Formal certificate. Lean constructs the genuine simple graphs, certifies every degree, defines the normalized operator from actual neighbor sums and constructs the complete eigenbasis via invertible coordinates. It verifies all eigenvectors and both multiplicity counts, forms the empirical Dirac measures, proves their exact constant value and their weak convergence in the probability-measure topology. The standard neighborhood support definition is written explicitly because this Mathlib pin has no packaged measure-support function. It verifies the zero-mass open neighborhood and refutes support equality with $[0, 2]$. Final audits use only standard Lean and Mathlib axioms.